Bluetooth Primer for RAG Experiments

A compact, self-written summary of Bluetooth Core Specification topics used as a fallback document.

1 Introduction and Scope

This primer is an educational summary of selected topics from the Bluetooth Core Specification. It is NOT the official specification. It exists so that a Retrieval-Augmented Generation (RAG) demo can run without downloading the full specification, which is several thousand pages long.

Bluetooth is a short-range wireless technology operating in the unlicensed 2.4 GHz Industrial, Scientific and Medical (ISM) band, from 2400 MHz to 2483.5 MHz. The Core Specification defines two main radio technologies: Basic Rate / Enhanced Data Rate (BR/EDR), often called Classic Bluetooth, and Bluetooth Low Energy (LE). A device that implements both is called a dual-mode device.

The architecture is split into a Host and a Controller. The Controller contains the radio, the baseband or link layer, and the link manager. The Host contains upper layers such as L2CAP, the Attribute Protocol (ATT), the Generic Attribute Profile (GATT), the Security Manager Protocol (SMP) and the Generic Access Profile (GAP). Host and Controller communicate through the Host Controller Interface (HCI), which defines commands, events and data packets over a transport such as UART or USB.

2 BR/EDR Radio

The BR/EDR radio uses 79 RF channels spaced 1 MHz apart, from 2402 MHz to 2480 MHz. Channel k has the centre frequency f = 2402 + k MHz, for k = 0 to 78.

Basic Rate uses Gaussian Frequency Shift Keying (GFSK) with a symbol rate of 1 Msym/s, giving a raw bit rate of 1 Mb/s. Enhanced Data Rate keeps the GFSK access code and header but switches the payload to phase shift keying: pi/4-DQPSK provides 2 Mb/s and 8DPSK provides 3 Mb/s.

A BR/EDR piconet consists of one Central (historically called master) and up to seven active Peripherals (historically called slaves). The Central defines the piconet clock and the frequency hopping sequence. In the connection state the radio hops at a nominal rate of 1600 hops per second, one hop per 625 microsecond time slot; multi-slot packets occupy 3 or 5 slots on a single frequency.

3 Transmitter Power Classes

Devices are classified by the maximum output power they can transmit. For BR/EDR the power classes are:

Power Class 1: maximum output power 100 mW (+20 dBm), minimum 1 mW (0 dBm). Power control is mandatory for Class 1 devices operating above +4 dBm.

Power Class 2: maximum output power 2.5 mW (+4 dBm), minimum 0.25 mW (-6 dBm).

Power Class 3: maximum output power 1 mW (0 dBm).

For Bluetooth Low Energy the specification defines the following LE power classes, all with a minimum output power of 0.01 mW (-20 dBm):

LE Power Class 1: maximum output power 100 mW (+20 dBm).

LE Power Class 1.5: maximum output power 10 mW (+10 dBm).

LE Power Class 2: maximum output power 2.5 mW (+4 dBm).

LE Power Class 3: maximum output power 1 mW (0 dBm).

Higher transmit power increases range but also increases current consumption and interference. Many LE devices therefore use transmit power control, adjusting the output power according to the received signal strength (RSSI) reported by the peer.

4 Adaptive Frequency Hopping (AFH) in BR/EDR

Adaptive Frequency Hopping was introduced in version 1.2 of the specification to improve coexistence with other systems in the 2.4 GHz band, most notably Wi-Fi. Instead of hopping over all 79 channels, a piconet using AFH hops only over channels classified as 'used', avoiding channels that suffer from interference.

AFH works in three steps. First, channel classification: each channel is classified as used, unused or unknown. The Central may measure channel quality itself, may request classification reports from Peripherals, and the Host may provide its own information with the HCI command HCI_Set_AFH_Host_Channel_Classification. Second, the Central communicates the resulting AFH channel map to the Peripherals with the LMP_SET_AFH message, which includes the instant at which the new map takes effect so that all devices switch simultaneously. Third, the hop selection kernel remaps any hop that would land on an unused channel onto one of the used channels.

The specification requires that at least N_min = 20 channels remain in the used set. This keeps the system a frequency-hopping spread spectrum system for regulatory purposes and preserves robustness. Channel classification is dynamic: as interference moves, the Central can update the map.

AFH does not change the hop rate. Packets on the same slot pair still follow the same-channel mechanism, where the Peripheral responds on the same frequency used by the Central in the preceding slot.

5 Bluetooth Low Energy Radio and PHYs

The LE radio uses 40 RF channels spaced 2 MHz apart, from 2402 MHz to 2480 MHz. Three of them are primary advertising channels: channel index 37 (2402 MHz), 38 (2426 MHz) and 39 (2480 MHz). They are placed to avoid the most commonly used Wi-Fi channels 1, 6 and 11. The remaining 37 channels (indices 0 to 36) are general purpose channels used for connections and secondary advertising.

LE defines three PHYs. LE 1M uses GFSK at 1 Msym/s and is mandatory. LE 2M, introduced in version 5.0, doubles the symbol rate to 2 Msym/s for higher throughput and shorter air time. LE Coded, also from version 5.0, uses the 1 Msym/s modulation with forward error correction: coding S=2 yields 500 kb/s and coding S=8 yields 125 kb/s, trading data rate for roughly two to four times the range.

6 LE Link Layer States and Connections

The LE Link Layer is described as a state machine. The basic states are Standby, Advertising, Scanning, Initiating and Connection. Later versions add the Synchronization state for periodic advertising and Isochronous Broadcasting states for LE Audio.

An advertiser transmits advertising packets on the primary advertising channels at an advertising interval between 20 ms and 10.24 s. A random delay advDelay of 0 to 10 ms is added to every advertising event so that two advertisers do not collide repeatedly.

A connection is created when an initiator responds to a connectable advertisement with a

connection request. Afterwards data exchange is organised into connection events. The connection interval is a multiple of 1.25 ms between 7.5 ms and 4.0 s. The Peripheral latency parameter allows the Peripheral to skip a number of consecutive connection events to save power when it has no data to send. The supervision timeout, between 100 ms and 32 s, is the maximum time between two received packets before the link is considered lost.

7 LE Channel Selection and Channel Maps

LE connections also hop between channels, once per connection event. Channel Selection Algorithm #1 uses a fixed hop increment between 5 and 16: unmappedChannel = (lastUnmappedChannel + hopIncrement) mod 37. If the unmapped channel is marked unused in the channel map, it is remapped onto the table of used channels.

Channel Selection Algorithm #2, introduced in version 5.0, derives a pseudo-random channel from the connection's Access Address and the connection event counter. It spreads traffic more evenly and is also used for periodic advertising and isochronous channels.

The Central maintains the channel map, classifying each of the 37 data channels as used or unused, and updates it with the LL_CHANNEL_MAP_IND control PDU, which contains the new map and the instant at which it applies. The Host can contribute with HCI_LE_Set_Host_Channel_Classification. In LE, a channel map shall contain at least two used channels. This is the LE equivalent of AFH.

8 Attribute Protocol and GATT

The Attribute Protocol (ATT) lets a client read and write attributes stored on a server. Each attribute has a 16-bit handle, a type identified by a UUID, a value and permissions. The default ATT_MTU on LE is 23 bytes; client and server can negotiate a larger MTU with the Exchange MTU procedure.

The Generic Attribute Profile (GATT) organises attributes into a hierarchy of services, characteristics and descriptors. A characteristic declaration is followed by its value attribute and optional descriptors such as the Client Characteristic Configuration Descriptor (CCCD), which a client writes to enable notifications or indications. Notifications are unacknowledged, while indications require a confirmation from the client.

9 Security: Pairing and Bonding

In LE, pairing is performed by the Security Manager Protocol (SMP). LE Legacy Pairing is the original method. LE Secure Connections, introduced in version 4.2, uses Elliptic Curve Diffie-Hellman (ECDH) on the P-256 curve to derive the Long Term Key (LTK), protecting against passive eavesdropping.

The association model is chosen from the devices' input and output capabilities: Just Works (no user interaction and no protection against man-in-the-middle attacks), Numeric Comparison (both devices display a six-digit number the user confirms; Secure Connections only), Passkey Entry (a six-digit passkey is typed on one or both devices) and Out of Band (keys are exchanged over another channel such as NFC).

Bonding means storing the keys produced by pairing so that future connections can be encrypted without pairing again. LE link encryption uses AES-CCM with a 128-bit key.

10 Glossary

AFH - Adaptive Frequency Hopping. ATT - Attribute Protocol. BR/EDR - Basic Rate / Enhanced Data Rate. CCCD - Client Characteristic Configuration Descriptor. GAP - Generic Access Profile. GATT - Generic Attribute Profile. HCI - Host Controller Interface. ISM - Industrial, Scientific and Medical band. L2CAP - Logical Link Control and Adaptation Protocol. LE - Low Energy. LMP - Link Manager Protocol. LTK - Long Term Key. MTU - Maximum Transmission Unit. PHY - Physical layer. RSSI - Received Signal Strength Indicator. SMP - Security Manager Protocol.